Install GNU Arm Embedded Toolchain

Steps:

1. Download GNU ARM Embedded toolchain
2. Install & setup Make
3. Install ST-Link Tools

    Download libusb driver

Make Project and Files

Main.c

Startup.s

Linker.ld

Makefile

Compile and flash

1. Download GNU Arm Embedded Toolchain

1. Go to the official ARM developer site: https://developer.arm.com/downloads/-/arm-gnu-toolchain-downloads
2. Download the version for **Windows (x86_64)**
   File name will look like: (Download .msi for windows)

   arm-gnu-toolchain-15.2.rel1-mingw-w64-i686-arm-none-eabi.msi

   After installing
   Important: add path to environment variable
   (C:\Program Files (x86)\Arm\GNU Toolchain mingw-w64-i686-arm-none-eabi\bin)
   Which contain files like:
   arm-none-eabi-xx
   Check installation:

File	Purpose
.msi	Windows installer (automatic setup)
.zip	Manual setup (you must configure PATH yourself)
.asc / .sha256	Verification files (not needed normally)

```
arm-none-eabi-gcc --version
```

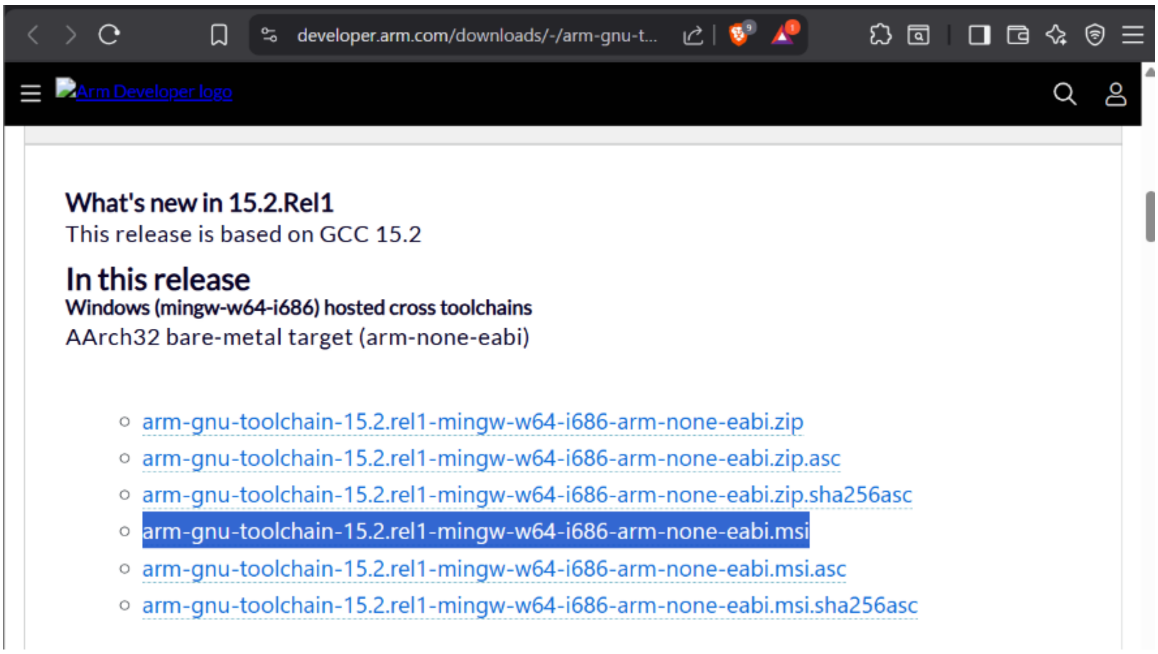

Figure 1arm embedded toolchain file for windows this toolchain consist of:

Tool	Purpose
arm-none-eabi-gcc	compile C code
arm-none-eabi-ld	linker
arm-none-eabi-objcopy	convert ELF → BIN
arm-none-eabi-gdb	debugging
arm-none-eabi-size	show program size

2. Install Make for Windows

Download: https://gnuwin32.sourceforge.net/packages/make.htm

Download complete package except source

Download

If you download the Setup program of the package, any requirements for running applications, such as dynamic link libraries (DLL's) from the dependencies as listed below under Requirements, are already included. If you download the package as Zip files, then you must download and install the dependencies zip file yourself. Developer files (header files and libraries) from other packages are however not included; so if you wish to develop your own applications, you must separately install the required packages.

Description	Download	Size	Last change	Md5sum
• Complete package, except sources	Setup	3384653	25 November 2006	8ae51379d1f3eef8360df4e674f17d6d
• Sources	Setup	1252948	25 November 2006	b896c02e3d581040ba1ad65024bbf2cd
• Binaries	Zip	495645	25 November 2006	3521948bc27a31d1ade0dcb23be16d49
• Dependencies	Zip	708206	25 November 2006	d370415aa924fa023411c4099ef84563
• Documentation	Zip	2470575	25 November 2006	43a07e449d4bab3eb3f31821640ecab7
• Sources	Zip	2094753	25 November 2006	8bed4cf17c5206f8094f9c96779be663

You can also download the files from the GnuWin32 files page.

You can monitor new releases of the port of this package.

Installation, Usage and Help

General Installation Instructions

GnuWin32 Help (feature requests, bugs, etc)

Requirements

Requirements for running applications, excluding external ones such as msvcrt.dll, perl, etc, are included in the Setup program and the dependencies zip file.

- Win32, i.e. MS-Windows 95 / 98 / ME / NT / 2000 / XP / 2003 with msvcrt.dll and msvcp60.dll. If msvcrt.dll or msvcp60.dll is not in your Windows/System folder, get them from Microsoft, or (msvcrt.dll only) by installing Internet Explorer 4.0 or higher.
- libintl3
- libiconv2

Check:

```
make --version
```

Add path to Environment Variables:

```
C:\Program Files (x86)\GnuWin32\bin
```

Verify Installation

Open **Command Prompt** and run:

make –version

3. Install ST-Link Tools

Download from the official repo:

👉 https://github.com/stlink-org/stlink/releases

Download the **Windows package**.

Look for something like:

stlink-1.x.x-win64.zip

Extract it somewhere, for example:

C:\stlink

Inside you will see:

st-info.exe
st-flash.exe
st-util.exe
copy “config” folder present in “stlink-1.8.0-win32\Program Files (x86)” into “C:stlink/”

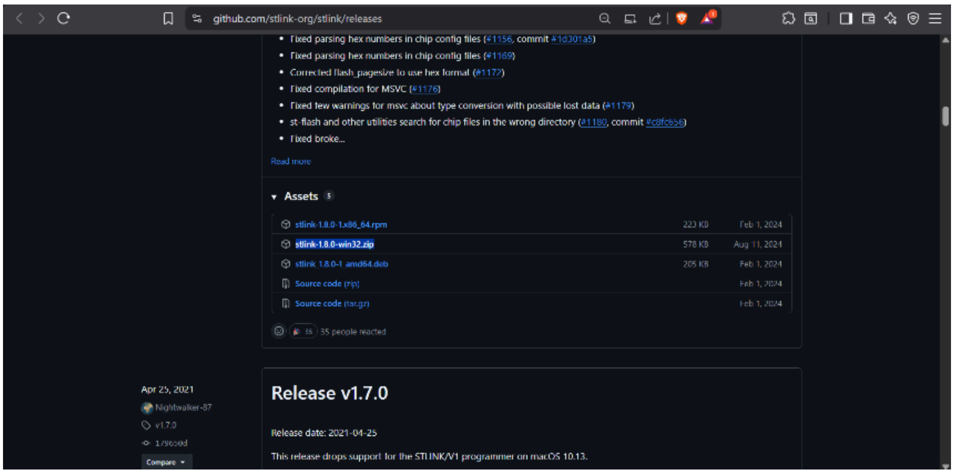

Figure 2 ST-link flash for windows

Download libusb (Important)

Download the Windows binary from:

https://libusb.info/

Figure 3 libusb Driver for windows

Download "latest windows Binaries"

After extracting the libusb

Open directory: "libusb-x.x/MinGW32/dll"

And copy libusb-1.0.dll to dir: "C:\....\stlink-1.8.0-win32\bin"

Now finally connect the development board and run command

.\st-info –probe

Into the dir.

This should show the

> Found 1 stlink programmers
> version: V2J46S33
> chipid: 0x431
>
> Output may vary with your development board.

Project Folder should contain:

Main.c

Startup.s

Linker.ld

Makefile

code: https://github.com/BhuneshSen/ARM-cortex-m/tree/main/led-blink-no-IDE

File	**Purpose**
main.c	Your program
startup.s	Boot code + vector table
linker.ld	Memory layout
Makefile	Compile & flash

Now open your project dir. And make

And run command:

C:\...\ stlink-1.8.0-win32\bin\st-flash write main.bin 0x08000000

Output

st-flash 1.8.0

Flash written and verified! jolly good!

Steps for compilation:

Open project folder run make

Then, C:\...\ stlink-1.8.0-win32\bin\st-flash write main.bin 0x08000000